

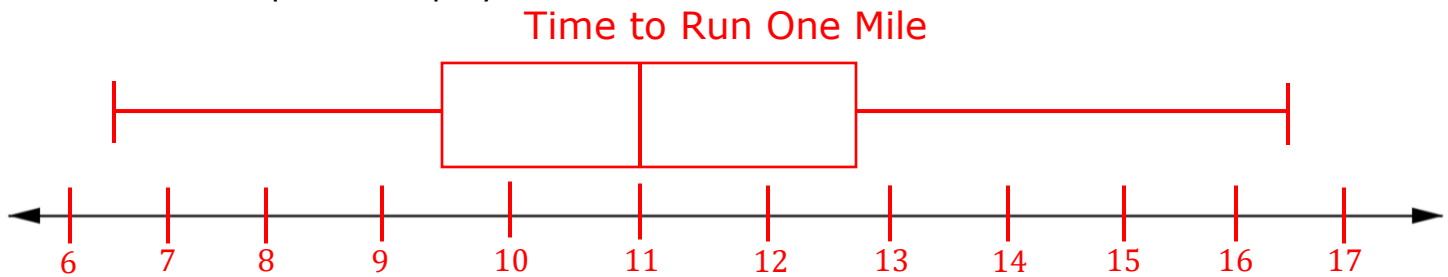
Orion collected data from students in his class to answer the question: "How many minutes does it take my classmates to run *one* mile?" His data are shown.

8.5	7	9.5	11.5	8	10	9.5	11	12.5	15
11	11.5	10	6.5	13	12.5	10.5	13.5	15	16.5

1. Determine the five-number summary for Orion's data.

Minimum: 6.5
First Quartile: 9.5
Median: 11
Third Quartile: 12.75
Maximum: 16.5

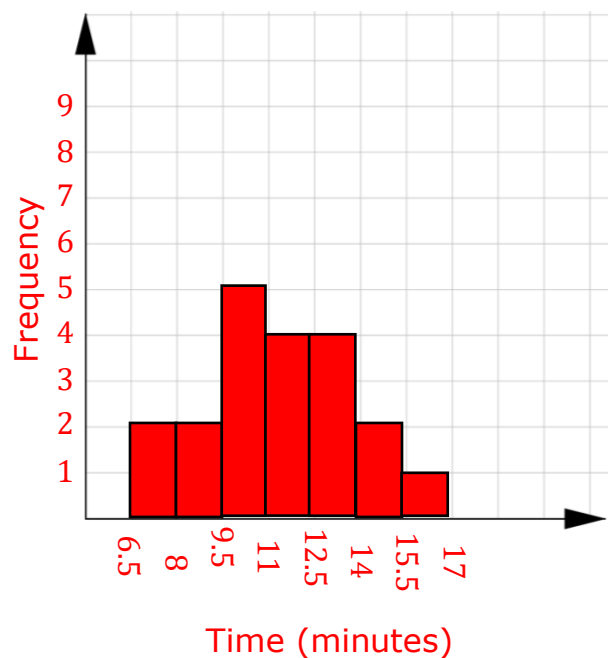
2. Create a box plot to display Orion's data.



3. Create a frequency chart of the data.

Time (min.)	Frequency
6.5 – 8	2
8 – 9.5	2
9.5 – 11	5
11 – 12.5	4
12.5 – 14	4
14 – 15.5	2
15.5 – 17	1

4. Create a histogram of the data.



5. What is the mean of Orion's data?
11.125

6. What is the median of Orion's data?

11

7. Describe the distribution's shape, center, spread, and unusual features.

The shape is a bell curve, mostly symmetrical, with one peak. The center is approximately the same as the mean and the median at 11.

8. Summarize Orion's results using statistical terminology.

The times to run a mile had a range of 10 minutes with the mean being 11.125 minutes to run one mile. Using the median, half of the students took less than 11 minutes to run the mile, and the other half took more than 11 minutes to run the mile.

9. Explain which type of graph, box plot or histogram, is better for displaying Orion's data.

The histogram is more useful at showing the amount of students who ran one mile at the different time intervals.

10. What could Orion have done to gather truthful data to answer his statistical question?

Answers may vary.

To get a better sample or larger sample size (like the whole 6th grade instead of just his class), he could have randomly chosen five students from each sixth grade gym class.